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Key Points:

- The South Qiangtang block was located at ~30°N in the Late Triassic
- In the Permo-Triassic the South Qiangtang block drifted rapidly northward at ~13.4 cm/yr
- The Longmuco-Shuanghu Ocean closed no later than ca. 222 Ma

Supporting Information:

Supporting Information may be found in the online version of this article.

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Paleomagnetism of Late Triassic Volcanic Rocks From the South Qiangtang Block, Tibet: Constraints on Longmuco-Shuanghu Ocean Closure in the Paleo-Tethys Realm

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Abstract The South Qiangtang block of the Qinghai-Tibet Plateau represents an area critical to understanding the late Paleozoic and early Mesozoic history of the Tethyan realm, but its drift history remains poorly constrained. Here we report a new quantitative paleogeographic constraint for the South Qiangtang block from a paleomagnetic study of Late Triassic volcanic rocks of the Xiaoqiebao Formation. A characteristic remanent magnetization isolated from 25 sites passes both fold- and reversal tests, and likely represents a primary magnetization. On the basis of these data, we estimate that the South Qiangtang block occupied a paleolatitude of 30.1 ± 4.6°N at ca. 222 Ma. When combined with existing paleomagnetic constraints, these new results indicate that the South Qiangtang block (and other "Cimmerian" blocks) moved rapidly northward (in true latitude) between the middle Permian and Late Triassic. Our new data further suggest that the southern branch of the Paleo-Tethys (Longmuco-Shuanghu Ocean) likely closed by the mid-Late Triassic.

Plain Language Summary The Paleo-Tethys was a major eastward-widening oceanic domain that separated eastern Gondwana and eastern Laurasia during Carboniferous-Permian time. The eventual disappearance of this ocean coincided with the amalgamation of the terranes comprising the modern Qinghai-Tibet Plateau. However, the plate kinematic history that led up to this suturing remains poorly constrained. In particular, the South Qiangtang block, which is thought to have formed the southern margin of the system, is a key area in need of additional constraints. In this work, we present new paleomagnetic results which indicate that the South Qiangtang block drifted rapidly northward between the middle Permian and Late Triassic (at an average south-north speed of ~13.4 cm/yr) to arrive to a paleolatitude of 30°N by 222 Ma. Such a position suggests that the southern branch of the Paleo-Tethys (Longmuco-Shuanghu Ocean) may have closed by this time.

1. Introduction

The Paleo-Tethys Ocean was a major eastward-widening ocean that divided eastern Laurasia and eastern Gondwana in Carboniferous-Permian time (e.g., Metcalfe, 2013; Stampfli et al., 2013; Torsvik & Cocks, 2013; Torsvik et al., 2012; Zhao et al., 2018). According to conventional paleogeographic models, in the latest Paleozoic and early Mesozoic the Paleo-Tethys progressively shrank due to the northward drift of a series of crustal blocks that rifted from the northern margin of Gondwana (the "Cimmerian" continent or terranes, including the Qiangtang block(s), Sibumasu, Iran block, Afghanistan, and others) (Angiolini et al., 2013; Song et al., 2017; Stampfli & Borel, 2002; Wan et al., 2019; Şengör et al., 1988), and the ocean finally closed by the collision and amalgamation of those terranes with the southern margin of Eurasia (Li et al., 2009, 2019; Pullen et al., 2008; Wang et al., 2018; Xu et al., 2015; Zhao et al., 2018). However, despite the attractiveness of this simplistic paleogeographic narrative, closure of the Paleo-Tethys was likely not a singular event, as multiple corridors proposed to be sutures crossing the Qinghai-Tibet Plateau have been related to the "Paleo-Tethys." Because the collapse of this once-massive oceanic domain is central to an understanding of the tectonic evolution of central Asia, as well as the regional environmental and biogeographic setting in the late Paleozoic and early Mesozoic, it is important to unravel the detailed internal kinematics of its demise.

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Proposed "Paleo-Tethys" sutures include the East Kunlun-Animaqing suture (EKAS) that broadly demarcates the southern margin of Tarim and Qaidam, the Xijinwulan-Jinshajiang suture (XJS) that forms the northern border of the North Qiangtang block (NQB) and the Longmuco-Shuanghu suture (LSS), which separates the NQB and South Qiangtang block (SQB), (e.g., Chang & Zheng, 1973; Kapp et al., 2000, 2003; Li, 1987; Liu et al., 2019; Li, Zhai, Dong, Zeng, & Huang, 2007; Metcalfe, 2013; Pan et al., 2004; Pullen et al., 2008; Yin & Harrison, 2000) (Figure 1a). These various sutures may represent different "branches" of the Paleo-Tethys, which should thus be treated as an oceanic realm rather than a singular basin. Following this, the EKAS and XJS represent a composite northern branch—the "Jinshajiang Ocean" (JSO), whereas the LSS is interpreted to mark a distinct southern branch—the "Longmuco-Shuanghu Ocean" (LSO) (Wu et al., 2020; R. X. Zhu et al., 2022).

Paleomagnetic and geologic data suggest that the JSO narrowed with the rapid northward drift of the NQB during the late Permian to Triassic, and closed by the collision of the NQB with the southern margin of Laurasia by the early Late Triassic (Cheng et al., 2023; Guan et al., 2021; Yu et al., 2022). However, the evolution of the LSO remains less clear (e.g., Li, 1987; Zhang, Zhang, et al., 2006; Y. C. Zhang et al., 2016; Yin & Harrison, 2000; Kapp et al., 2003; Gehrels et al., 2011) and proposals of its closure time range from the late Permian to the Late Triassic (e.g., Qi et al., 2009; Xie et al., 2018; Yang et al., 2011; Zhai et al., 2018). As the SQB represents the southernmost block of the Qinghai-Tibet Plateau that once bounded the LSO, constraints on the latest Paleozoic to early Mesozoic drift history of this block are needed to constrain the evolutionary collapse of the LSO, and thus the Paleo-Tethys realm more broadly. However, late Paleozoic-early Mesozoic quantitative paleomagnetic constraints on the SQB are presently very sparse (Wei et al., 2022).

Here we report on new paleomagnetic data collected from Late Triassic volcanic rocks from the SQB. Combined with existing Permian paleomagnetic data, we apply these new data to further constrain the drift history of the SQB during the latest Paleozoic-early Mesozoic, in order to further constrain the kinematic evolution of Paleo-Tethys closure.

2. Geological Setting and Sampling

The sampling area of this study is located in the centeral part of the northern margin of the SQB, 50 km south of the LSS, near Xiaochaka Lake and Pengyan Co Lake, in Shuanghu County (Figure 1b). In this area, Late Triassic strata are widespread and observed to overlie late Carboniferous rocks with angular unconformity. Contacts between Late Triassic strata and Permian and Jurassic strata are exclusively faulted (Figure 1b). The Late Triassic successions comprise, from bottom to top: the Xiaoqiebao Fm. (also called the Nadigangri Fm. by Li (2019)), Jiangzhong Fm., Jiaomuchaka Fm. and Zhana Fm. (Figure 1b). The Jiangzhong Fm., Jiaomuchaka Fm., and Zhana Fm. are mainly composed of limestone and sandstone, whereas the Xiaoqiebao Fm. is a bimodal volcanic sequence (Li, 2019). The latter is mainly comprised of grayish green or purple amygdaloidal basalt, basaltic andesite, basaltic breccia, and rhyolite. Limestones have also been reported to occur among the volcanics of the Xiaoqiebao Fm. (Feng et al., 2005), but we did not observe any interlayered limestones in the sampling area. Geochemically, the Xiaoqiebao volcanics have been associated with a rift-related intra-plate extensional environment (Fu et al., 2010a, 2010b; Li, 2019; Wang et al., 2022). A K-Ar date of 223 ± 5 Ma is reported from volcanic rocks of the Xiaoqiebao Fm. in the study area (Wang, Qu, et al., 2007; Figure 1b), and Li (2019) reported zircon U-Pb ages of 221.9 ± 3.4 Ma (MSWD = 1.6) near the Xiaochaka lake and 221.8 ± 2.1 Ma (MSWD = 0.8) from the Jianshishan section. We thus calculate the average age of the Xiaoqiebao Fm. as 222.0 ± 3.3 Ma. A Late Triassic age is further supported by sporopollen (*Schizosporites* cf. *parvus*, *Psophosphaera bullulinaeformis*, *Cycadopites* sp., *Chasmatosporites* sp., *Megamonoporites cacheutensis*, *Megamonoporites* sp.) and bivalve (*Halobia* sp., *Burmesia* sp., *Chlamys* sp.) fossils found among limestone interlayers of the Xiaoqiebao Fm. (Feng et al., 2005). The bottom boundary of the Xiaoqiebao Fm. in the study area is a paleo-weathering crust (Fu et al., 2009; Wang, Fu, et al., 2007; Wang et al., 2022) that regionally mantles the Carboniferous-Permian strata (Fu et al., 2009, 2010a; Zhu et al., 2005).

Structurally, the study area exhibits clear evidence of several episodes of Mesozoic and Cenozoic tectonism. The pre-Late Triassic angular unconformity capped by a paleo-weathering crust, together with the absence of Early and Middle Triassic strata, indicates that the region was deformed and uplifted in Early to Middle Triassic time (Wang & Fu, 2018). The Late Triassic and Jurassic strata in the study area mostly strike ESE-WNW and dip both to the SSW and NNE. In the northeast of the study area, near Kongkongchaka Lake, the Late Triassic strata are disconformably overlain by Early-Middle Jurassic strata (Wang et al., 2022). The occurrence of another

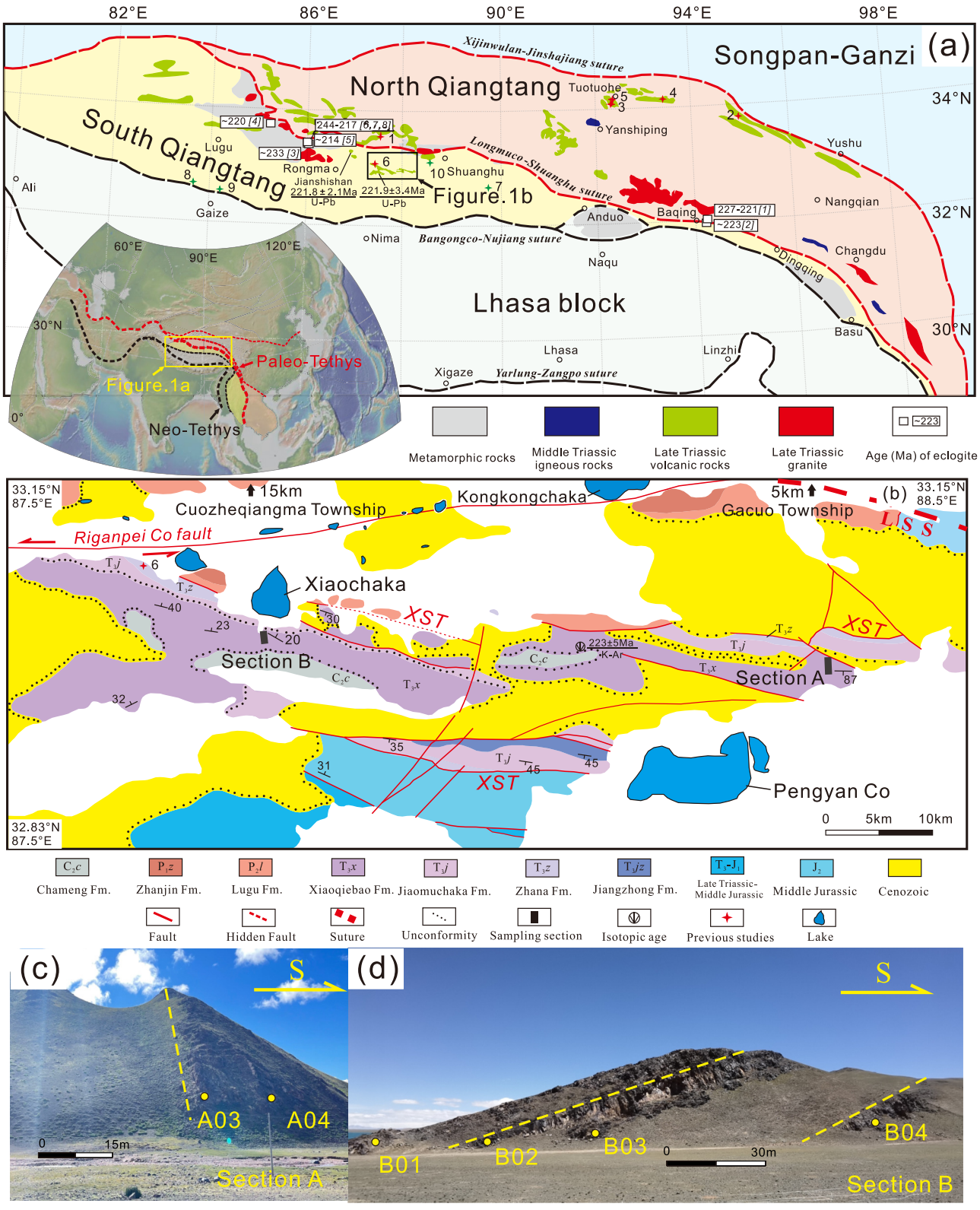


Figure 1.

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angular unconformity between Jurassic and Cenozoic strata reflects another episode of compressional deformation and regional uplift that is ascribed to the collision of Lhasa against the SQB (marking the closure of the Bangongco-Nujiang Ocean) in late Early Cretaceous to early Late Cretaceous time (Liu et al., 2001, 2002; Wu et al., 2011, 2012; Ren et al., 2016; Li et al., 2017; Zhao et al., 2019a, 2019b; Cao et al., 2020; Ma et al., 2023). Subsequently, during the Cenozoic Himalayan orogeny, the collision of India against Eurasia resulted in another strong compressional episode in the SQB, and the development of large-scale WNW-trending thrust structures (e.g., Xiaochaka-Shuanghu thrust) and transcurrent structures (e.g., Riganpei Co left-lateral strike-slip fault) (Figure 1b; Liu et al., 2022).

Paleomagnetic samples were collected from two sections, here called Section "A" (33°N, 88.4°E) and Section "B" (33°N, 87.8°E) (Figure 1b). In Section A, the volcanic rocks dip ∼87° to the south, whereas in Section B the volcanic rocks dip 17–28° to the north. In Section A, 114 samples from 13 sites were drilled from grayish green basalt, purplish red basaltic andesite and red rhyolite (Figure 1c). In Section B, 91 samples from 12 sites were drilled from grayish green basalt. In both sections, a "site" represents a distinct volcanic flow (Text S1 in Supporting Information S1), which could be distinguished by lithological and textural variations (e.g., quenched margins; Figures S1a and S1b in Supporting Information S1), and paleohorizontal was estimated from flow structures (Figures S1c–S1f in Supporting Information S1). Samples were collected using a gasoline-powered drill and were oriented in situ using magnetic and solar compasses (yielding an uncertainty of <5°). A handheld GPS unit was used to determine the geographic coordinates of the sampling sites, and structural orientations were measured with a magnetic compass.

3. Results

Detailed rock-magnetic experimental methods, their results and scanning electron microscopy observations are presented in Text S2 in Supporting Information S1, and paleomagnetic methods are presented in Text S3 in Supporting Information S1; in the following we summarize the key results. On the basis of our rock magnetic experiments, we interpret both magnetite and hematite to be the main magnetic carriers in the samples from Section A, whereas in Section B the main magnetic carrier is magnetite alone.

In section A, the demagnetization curves of most specimens typically exhibit two-component behavior: a low-temperature remanent component (LTC) was removed by approximately by 350°C, and a high-temperature component (HTC) was isolated between 350 and 585°C or 350 and 680°C (Figures 2a and 2b); we deem the HTC the characteristic remanent component (ChRM). In some specimens little decay occurred below 350°C, and only the high-temperature ChRM could be defined (Figure 2c). In section B, all specimens display two components of magnetization: a LTC removed below 350°C gives way to a high-temperature component that decays univectorially to the origin between 350 and 585°C (Figures 2d). Although we found thermal demagnetization (TD) to be most effective, experiments with alternating field demagnetization (AF) and hybrid (AF + TD) treatments yielded results consistent with those described above (Figures 2e and 2f). The ChRM data of the individual samples are provided in Table S1 (Wei, 2023).

The directions of the LTC from both sections are similar, and a mean calculated from 161 samples is $D_g = 356.4°$, $I_g = 51.1°$, $k_g = 9.8$, $\alpha_{95g} = 3.7°$ in geographic coordinates and $D_s = 12.5°$, $I_s = 82.9°$, $k_s = 1.9$, $\alpha_{95s} = 11.8°$ after tilt correction. In geographic coordinates, the mean direction is close to the direction of the present geomagnetic field ($D = 0.7°$, I = 51.9). Furthermore, the dispersion of the LTC directions increases significantly with tilt correction (Figures 2g and 2h). Stepwise tectonic correction of the LTC (Watson & Enkin, 1993) yields an optimal clustering at $-11.4 \pm 1.5\%$ unfolding, indicating that the magnetization post-dates folding and is thus a remagnetization (Figure 2i).

In geographic coordinates, the ChRM directions from both sections are south-pointing, but have inclinations of opposite sign (Figure 2j). The mean ChRM direction computed from the 13 Section A sites is $D_g = 196.6°$,

Figure 1. (a) Simplified regional geological map showing the locations of the Triassic igneous rocks in Qiangtang (modified after Li et al. (2019) and He et al. (2022)). Red crosses depict the location of Triassic paleomagnetic studies: 1-Song et al. (2012); 2-Yu et al. (2022); 3-Song et al. (2015); 4-Lin and Watts (1988); 5-Zhou et al. (2019); 6-Song et al. (2012). Green crosses depict the locations of Jurassic-Cretaceous paleomagnetic studies: 7-Cao et al. (2019); 8-Cao et al. (2020); 9-Chen et al. (2017); 10-Meng et al. (2018). White squares are the age of eclogite: 1-Jin et al. (2019); 2-X. Z. Zhang et al. (2018); 3-Dan et al. (2018); 4-Zhai et al. (2011); 5-Zhang et al. (2010); 6-Zhai et al. (2017); 7-Pullen et al. (2011); 8-Pullen et al. (2008). (b) Regional geological map showing the sampling section (modified after Jiangaidarina, and Paduco regional geological survey reports). LSS: Longmuco - Shuanghu Suture; XST: Xiaochaka-Shuanghu thrust. (c–d) Photographs of the Triassic volcanic rocks in Section A and Section B.

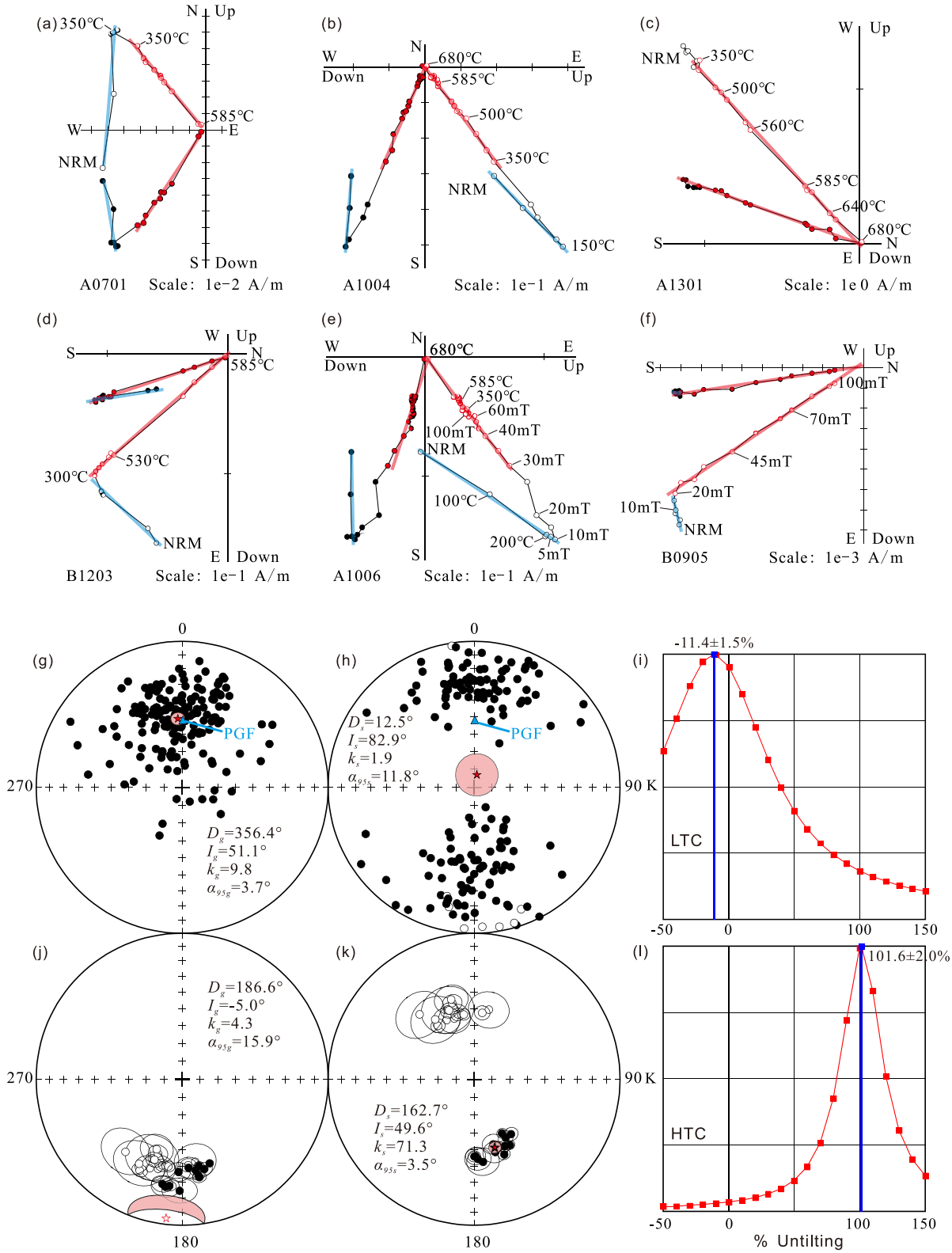


Figure 2.

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$I_g = -40.1°$, $k_g = 60.4$, $\alpha_{95g} = 5.4°$ in geographic coordinates and $D_s = 342.2°$, $I_s = -50.3°$, $k_s = 60.1$, $\alpha_{95s} = 5.4°$ after tilt correction, and from the 12 section B sites is $D_g = 176.6°$, $I_g = 33.7°$, $k_g = 70.0$, $\alpha_{95g} = 5.2°$ in geographic coordinates and $D_s = 163.2°$, $I_s = 48.8°$, $k_s = 81.9$, $\alpha_{95s} = 4.8°$ after tilt correction. With tilt-correction, Section A directions become north-pointing, approximately antipodal to those of Section B. Accordingly, after tilt-correction, we find that the ChRMs of sections A and B pass the McFadden and McElhinny (1990) and Heslop and Roberts (2018) reversal tests (Table S2; Wei, 2023). The execution of several fold tests on these 25 ChRMs (after inverting them to a common polarity) also yielded positive results at the 99% confidence level, with optimal clustering occurring at 101.6 ± 2.0% unfolding (Figure 2l and Table S2; Wei, 2023).

Given the results of the fold and reversal tests, which establish that the ChRMs of the two sections share a common mean and pre-date folding, we merge the directions from the 25 sites and calculate a mean, tilt-corrected direction of $D_s = 162.7°$, $I_s = 49.6°$, $k_s = 71.3$, $\alpha_{95s} = 3.5°$ (Figure 2k). An alternative mean computed from 199 specimen-level directions is $D_s = 162.8°$, $I_s = 49.7°$, $k_s = 37.6$, $\alpha_{95s} = 1.6°$, which is not significantly different than the mean computed from site level directions. After transforming each of the 25 site means into virtual geomagnetic poles (VGPs), we compute a corresponding paleomagnetic pole at $\lambda = -24.4°$N, $\varphi = 104.5°$E, $A_{95} = 4.3°$. With respect to theoretical expectations for paleosecular variation (PSV), this pole meets the N-specific criterion of Deenen et al. (2011) (critical A_{95} values of 3.5–6.9° for $N = 25$), which suggests that PSV has largely been averaged out.

4. Discussion

4.1. Age of the Magnetization and the Paleolatitude of the SQB During Late Triassic

The result of the fold test clearly demonstrates that the age of the ChRM isolated from the Xiaoqiebao Fm. volcanic rocks pre-dates the age of the folding of those rocks. The most recent deformation episode to have affected the Triassic strata can be associated with the Cenozoic collision of India against Eurasia (Zhao et al., 2019, 2019a, 2019b), which at least constrains the timing of ChRM acquisition to the pre-Paleogene. However, the angular unconformity between the Jurassic and Cenozoic strata (and the general sparsity of Cretaceous strata in the region) indicates that significant deformation also occurred during the earlier Cretaceous collision of Lhasa against the SQB, and so we may further infer that the ChRM is pre-Cretaceous in age. Given that the ChRM is associated with antipodal directions held in different minerals from several lithologies, we contend that the magnetization is most likely primary.

The only other Late Triassic paleomagnetic constraint from the SQB is from the Zhana Fm. sandstone, as reported by Song et al. (2012), but this constraint has a significant shortcoming. Being derived from clastic sedimentary rocks, the paleomagnetic inclinations may have been flattened, but Song et al. (2012) did not conduct any tests to evaluate this potential bias. In this regard it is worth noting that our mean paleomagnetic direction is steeper than that of Song et al. (2012) (Figures 3a and 3b). Owing to this important but unquantified potential bias, we do not further consider this constraint.

The position of our new pole differs from existing post-Late Triassic poles reported from the SQB (Figure 3a; Text S4 in Supporting Information S1). Normally this could be used as an additional argument against remagnetization, but these younger paleomagnetic poles are distributed about a small circle centered on the sampling area, which suggests that significant vertical-axis rotations have occurred within the SQB since the Late Triassic (Figure 3a). Meng et al. (2018) likewise concluded from paleomagnetic data that a clockwise vertical-axis rotation of 57.3 ± 3.9° (relative to stable Eurasia) occurred in the Shuanghu basin since mid-Cretaceous time. However, note that even paleomagnetic data of similar age from the same region appear to have been significantly rotated relative to one another (e.g., Chen et al., 2017). Such rotations could have been caused by the Lhasa-SQB collision (e.g., Kapp et al., 2007; Liu et al., 2017; Hu et al., 2022; Z. C. Zhu et al., 2022) during the Cretaceous, but the dispersion of Late Cretaceous poles and the occurrence of large scale strike-slip faults associated with the India-Eurasia collision indicates this latter event was also a major contributor (e.g., Ding et al., 2005, 2022; Hu

Figure 2. (a–f) Representative demagnetization results of samples from the Xiaoqiebao Formation (shown in geographic coordinates). The colored circles and colored straight lines represent the points involved in the PCA and the fitted direction, respectively. Blue (red) corresponds to the low-temperature components (high-temperature components). Equal-area projection of the low-temperature component and high-temperature component before (g and j) and after (h and k) tilt correction. The blue triangle is the mean direction of the local present geomagnetic field (PGF) and the red star within the pink circle is the mean direction and its α_{95}. Open (filled) circles/stars are projections onto the upper (lower) hemisphere. (i and l) Partial untilting test (Watson & Enkin, 1993) of LTC and HTC. The red line shows the precision parameter k as a function of untilting, and the blue line marks the percent untilting that maximizes the precision parameter.

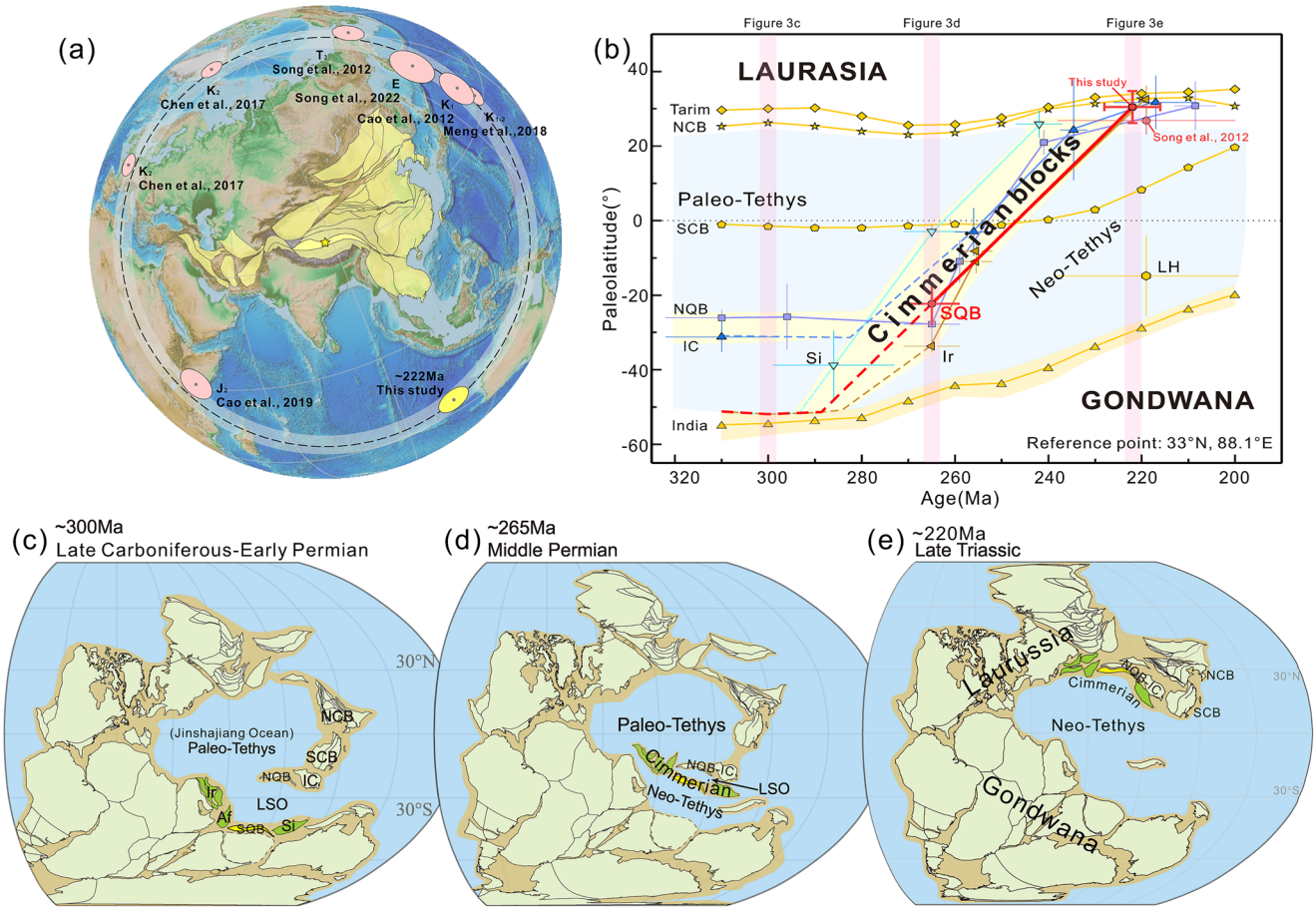


Figure 3. (a) Plot of post-Triassic paleomagnetic poles from SQB. Yellow star: sampling location; circles: Mesozoic paleomagnetic poles of SQB (Table S3 in Wei (2023)); light gray band: the error bounds of our new pole as a small circle centered on sampling site; (b) Paleolatitude evolution of the SQB and its surrounding blocks during the early Permian to Late Triassic (modified after Wei et al. (2022); the paleolatitude evolution of the North Qiangtang Block is modified after Cheng et al. (2023); see Table S3 for detailed paleomagnetic data). (c) Paleogeographic reconstructions of the SQB during the Early Permian to Late Triassic (modified after Huang et al., 2018; Yan et al., 2019; Wei et al., 2022).

et al., 2016; Jin et al., 2018; Suo et al., 2022). Because the sampling area may have suffered vertical-axis rotations, the declination of our reported ChRM should not be used for paleogeographic reconstruction, and we rely only on the inclination. The inclination indicates that the SQB was located at a latitude of 30.1 ± 4.6° at ~222 Ma.

4.2. The Permian to Triassic Drift History of SQB

Several lines of geological evidence suggest that the SQB was proximal to Sibumasu and the northern margin of Gondwana in the Permo-Carboniferous, including the occurrence of glacial relics (e.g., Li, 1987; Wang et al., 2021) and the provenance (age distribution and isotopic signature) of detrital zircons (e.g., Fan et al., 2015; Gao et al., 2022). An early Permian paleomagnetic result from Sibumasu places it at 40°S (Huang & Opdyke, 1991; Xu et al., 2015), which is consistent with it being positioned along the northern margin of Gondwana then (Figure 3c). By contrast, paleomagnetic constraints from Indochina (IC) and the North Qiangtang block (NQB) indicate that they occupied distinctly lower latitudes in the Permo-Carboniferous (~20°S; Cheng et al., 2012; Song et al., 2017; Yang et al., 2017; Yan et al., 2020; Cheng et al., 2023), as also reflected by their very different (warm water) faunas (Y. C. Zhang et al., 2016). On the basis of these sparse paleomagnetic constraints, we estimate that the width of the LSO (between SQB-Sibumasu and IC-NQB) was on the order of ~2,000 km at this time, while the JSO (between IC-NQB and Tarim-North China Block) exceeded 5,000 km (Figure 3c). Thus, the latter can most appropriately be considered the "main" Paleo-Tethys Ocean, although the LSO may have provided an important oceanic gateway connecting the Tethyan and Panthalassic realms.

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In the early Permian, a large-scale episode of dyke emplacement occurred along the northern margin of Gondwana, which may have been associated with a mantle plume (Zhai et al., 2013a; Y. X Zhang et al., 2018; Wang et al., 2019; Dan et al., 2021). In turn, that magmatism may have facilitated the disintegration of the northern margin of Gondwana by the breakout of numerous crustal blocks (Wang et al., 2019; Zhai et al., 2013a) that were already subject to tension imparted by the northward subducting Paleo-Tethys (Wan et al., 2019, 2021; Wu et al., 2020; R. X. Zhu et al., 2022; Zhu et al., 2023). The rifting of those crustal blocks (the "Cimmerian" blocks) from the northern margin of Gondwana marked the opening of the Neo-Tethys Ocean, which proceeded to widen as the crustal blocks drifted north. Northward migration of the SQB at this time is supported by its assemblage of middle Permian fossils, which reflect a transition from Gondwanan to Cathyasian faunal affinities (Shen et al., 2016; Zhang et al., 2014). More quantitatively, a middle Permian paleomagnetic result from the SQB places it at ~22°S then (Wei et al., 2022). Contemporaneous paleomagnetic data from the IC-NQB suggest they remained at low latitudes during the middle Permian, and so the LSO would have narrowed (as the Neo-Tethys grew), while the JSO remained wide (Figure 3d).

Our new paleomagnetic results indicate that in the Late Triassic the paleolatitude of the SQB was 30.1 ± 4.6° (Figure 3b). Although these data do not constrain the hemisphere, geological evidence that the NQB and SQB amalgamated in the Late Triassic (see next section) would suggest that the SQB laid at 30°N at this time (provided that the paleolatitude of NQB was ~32°N by ~227–222 Ma; Yu et al. (2022)). Comparing this ~222 Ma paleolatitude constraint with the ~265 Ma constraint (~22°S; Wei et al., 2022) allows the determination that the SQB must have drifted northward at a minimum average rate of 13.4 cm/yr between those times (Figure 3b). Middle Permian to Late Triassic paleomagnetic constraints from the Iranian block, another of the Cimmerian blocks, reflect a similarly rapid rate of northward drift (10–14 cm/yr) during this interval (Besse et al., 1998), when the NQB-IC also experienced rapid northward movement (Cheng et al., 2023; Ma et al., 2019; Yan et al., 2019; Zhou et al., 2019) (Figure 3b). Hence, this was a dynamic interval that witnessed swift plate motions and the rapid expansion of the Neo-Tethys Ocean at the expense of the Paleo-Tethys.

4.3. Ocean Closures in the Paleo-Tethys Realm

The main cratons of the East Asia continental collage (e.g., Tarim, North China Block and South China Block) are thought to have mostly assembled and collided with the north Eurasian body of the Pangea supercontinent by the mid-Late Triassic (Huang et al., 2018; Zhao et al., 2018). Late Triassic paleomagnetic data from the central-eastern NQB (Song et al., 2015; Zhou et al., 2019), eastern NQB (Yu et al., 2022) and northern IC (Yan et al., 2019) likewise suggest that the IC-NQB collided with the southern margin of the East Asian collage about the same time, at ~230 Ma (Yu et al., 2022). This implies that the northern branch of the Paleo-Tethys (the JSO) closed during the early Late Triassic (Figure 3e).

The existence of a southern branch of the Paleo-Tethys (the LSO) is supported by geologic evidence including Late Devonian radiolarian fauna (Li et al., 2024) and Cambrian to Permian age ophiolitic mélange exposed along the LSS (e.g., Hu, Li, Wu, et al., 2014; Zhai, Jahn, Wang, et al., 2013; Zhai et al., 2016; X. Z. Zhang et al., 2016; T. Y. Zhang et al., 2018), but the timing of its closure remains unclear. Previous proposals of LSO closing time span a range of intervals: late Permian—Early Triassic (Qi et al., 2009; Yang et al., 2011), Early Triassic (Xie et al., 2018), Middle Triassic (Hu, Li, Li, et al., 2014; Peng et al., 2015; Pullen et al., 2008; Wang et al., 2011) and Late Triassic (Dan et al., 2018; Dong et al., 2013; Li, Zhai, Dong, Yu, & Huang, 2007; Song et al., 2015; Zhai et al., 2011, 2018).

Our new paleomagnetic paleolatitude constraint supports the notion that the NQB and SQB merged into a unified whole by at least the mid-Late Triassic, given that they occupied similar latitudes by then (Figure 3b). Notably, this inference is based on the simplistic premise that the NQB and SQB occupied the same longitude at that time. It is therefore important to consider this interpretation (that the Paleo-Tethys ocean had closed at least by mid-Late Triassic time) against the geologic evidence. In the following we briefly review some key regional stratigraphic, metamorphic and geochemical observations.

Starting with a stratigraphic perspective, a key observation is that Early-Middle Triassic strata are absent across large areas of the SQB (including our sampling area), and the Late Triassic strata rest atop late Paleozoic rocks above a regional angular unconformity. This unconformity suggests that the SQB was uplifted and denuded in the Early to Middle Triassic (Wang & Fu, 2018), possibly resulting from terrane collision associated with Paleo-Tethys closure (Wang et al., 2022; Yan et al., 2016; Zhai et al., 2011). Similarly, in the middle of the LSS,

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unmetamorphosed continental sediments of the ca. 214 Ma Wanghuling Fm. rest with angular unconformity above late Paleozoic metamorphic rocks and ophiolite remnants (Li, Zhai, Dong, Yu, & Huang, 2007). The Wanghuling Fm. is recognized as the oldest continental sedimentary sequence covering the ophiolitic mélange in the LSS, which thereby places a minimum age on LSO closure.

The occurrence of high-pressure/low-temperature (HP-LT) metamorphic rocks in the LSS also provide important information about the closure of the LSO by way of the corresponding continental collision. Along the LSS, Late Triassic HP-LT metamorphic rocks form a >500 km long and up to 100-km wide belt that was generated by the partial subduction of oceanic crust and associated metasedimentary rocks during terminal closure of the LSO (Figures 1a and 3d; e.g., Kapp et al., 2000; Zhang, Cai, et al., 2006; Zhang, Fan, et al., 2018; Li, 2008; Dong & Li, 2009; Li et al., 2009; Zhang & Tang, 2009; Zhang et al., 2010; Zhai et al., 2011; X. Z. Zhang et al., 2018; Jin et al., 2019). More specifically, the peak (eclogite-facies) HP metamorphic stage interpreted to closely post-date terminal LSO closure has been estimated to ca. 227–223 Ma, whereas the subsequent cooling stage recording exhumation of the metamorphic rocks during post-collisional extension has been estimated to ca. 223–214 Ma. (Dan et al., 2018; Tang & Zhang, 2014; X. Z. Zhang et al., 2018; Y. X. Zhang et al., 2018; Zhai et al., 2011, 2017). These constraints again suggest that the LSO had closed no later than the mid-Late Triassic (Figure 3e).

A Late Triassic closure of the LSO is also supported by some geochemical evidence. During the Late Triassic the LSS was extensively intruded by granites that yield geochemical signatures interpreted as "post-collisional" (Figure 1a; e.g. Dong et al., 2013; Fu et al., 2010a, 2010b; Hu et al., 2010, 2014a; Li et al., 2015; Ma et al., 2016; Wang et al., 2015; Peng et al., 2015; Qian et al., 2016; Shi et al., 2012). Similarly, Late Triassic bimodal volcanic rocks emplaced into the SQB and NQB (e.g., Xiaoqiebao Fm. and 200–230 Ma Nadi Kangri Formation) have been interpreted to reflect an intra-plate extensional environment (Fu et al., 2010a, 2010b; Li, 2019; Wang et al., 2022), possibly associated with post-collisional extension. These inferences too would suggest that the LSO had closed at least by Late Triassic time (Figure 3e).

In summary, geological observations support the inference that the LSO had closed at least by mid-Late Triassic time (if not earlier), as suggested by paleomagnetic constraints. A more specific estimate of the closure timing of the LSO and a better understanding of its mode of closure (e.g., diachronic closure from east to west or west to east; e.g., Tang & Zhang, 2014; Lu et al., 2019; Xu et al., 2020; Y. X Zhang et al., 2018) will require additional paleomagnetic constraints from the SQB.

5. Conclusions

We have reported new paleomagnetic results from a study of Late Triassic volcanic rocks of the SQB. A ChRM derived from 25 sites passes the fold test and reversal test and likely represents a primary magnetization. These new paleomagnetic data indicate that the SQB was located at 30.1°N ± 4.6° at ~222 Ma. A consideration of these data together with other paleomagnetic and geologic constraints from neighboring continental blocks indicates that both the northern (Paleo-Jinshajiang Ocean) and southern (Longmuco-Shuanghu Ocean) branches of the Paleo-Tethys Ocean closed at a similar time, and no later than the mid-Late Triassic (~222 Ma).

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Data Availability Statement

Data used in this study (Tables S1 to S3) can be accessed at Zenodo (Wei, 2023).

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